



POWER BI MAJOR CAPSTONE PROJECT

END TO END

POWER BI PROJECT GUIDE

PROJECT TITLE

SCHOOL INFRASTRUCTURE & ENROLLMENT
ANALYTICS DASHBOARD

PROJECT CATEGORY

Education Analytics
School Analytics
Infrastructure Analytics

INSTITUTE NAME

Infinity Code Nexus

TRAINER / GUIDE

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CREATED BY

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TOOLS USED	Power BI Desktop Power Query, DAX & Power BI Service
SUBMISSION DATE	[10/06/2026]

| Prepared as part of the Power BI Masterclass Program to demonstrate skills in Data Analysis, Data Modeling , DAX Development, Dashboard Design, UI/UX, Storytelling, and Business Intelligence Reporting. | |

PROJECT OVERVIEW

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PROJECT OBJECTIVE

Develop a comprehensive Power BI dashboard to analyze school infrastructure, enrollment, teacher availability, and educational distribution across locations.

Participants are expected to:

- ✓ Analyze school infrastructure
- ✓ Evaluate student enrollment
- ✓ Assess teacher availability
- ✓ Compare school categories
- ✓ Generate meaningful insights
- ✓ Present recommendations

PROJECT GOAL

Convert raw school data into actionable insights that help understand educational facilities, student distribution, and school performance across different regions.

BUSINESS PROBLEM

Educational departments and school administrators need visibility into school infrastructure, student enrollment, and resource allocation.

The objective is to identify:

- Which regions have more schools?
- Which schools have higher enrollment?
- How are teachers distributed?
- Whether infrastructure is adequate?
- How school categories vary by region?

BUSINESS QUESTIONS

Participants must answer:

1. How many schools are available?
2. Which districts have more schools?
3. Which school categories dominate?
4. What is the total student enrollment?
5. What is the teacher-to-student ratio?
6. Which schools have better infrastructure?
7. Which regions need improvement?

DATASET INFORMATION

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DATASET NAME : Basic Details of Schools

DATASET DETAILS : Approximate Records 500+

Columns : 44

EXPECTED ANALYSIS AREAS

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School Analytics

Enrollment Analytics

Teacher Analytics

Infrastructure Analytics

Location Analytics

POWER QUERY TRANSFORMATIONS

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DATA CLEANING ACTIVITIES

- ✓ Validated School Records
- ✓ Handled Missing Values
- ✓ Corrected Data Types
- ✓ Standardized State Names
- ✓ Standardized District Names
- ✓ Verified Student, Teacher and Classroom Counts
- ✓ Removed Duplicate Records

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DATA PREPARATION ACTIVITIES

- ✓ Renamed Columns for Better Readability
- ✓ Replaced Blank and Null Values
- ✓ Corrected Data Formats
- ✓ Removed Unnecessary Columns
- ✓ Validated Numerical Columns
- ✓ Created Relationships Between Tables
- ✓ Prepared Data for Dashboard Analysis

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CUSTOM COLUMNS CREATED

- ✓ Teacher-Student Ratio
- ✓ Average Students Per School
- ✓ Average Teachers Per School

✓ Infrastructure Status Category

✓ School Location Classification (Urban/Rural)

SCREENSHOT

FileHomeHelpTable tools

Share

NameBasic Details of Sch...

Manage relationships

New measure

Quick measure

New column

New table

Mark as date table

Calendars

State (state_name)	State Code (state_code)	District (district_name)	District Code (district_code)	Sub-District (subdistrict_name)	Sub-District Code (subdistrict_code)	Cluster (cluster_name)
Telangana	36	Suryapet	696	Huzumagar	5333	Zpzs Lingagiri
Andhra Pradesh	28	Kurnool	511	Halaharvi	5167	Zpzs Gulyam
Maharashtra	27	Gondia	476	Deori	4385	Kakodi
Madhya Pradesh	23	Dindori	404	Amarpur	3829	Hss Amarpur (23410804816)
Assam	18	Bongaigaon	281	Srijangram	2506	Pachania
Karnataka	29	Vijayapura	530	Muddebihal	5796	Ingalagi
Maharashtra	27	Bid	0	Beed	0	V P Neknoor
Manipur	14	Kangpokpi	712	Kangpokpi	2370	Maram Junior High School
Bihar	10	Muzaffarpur	208	Aurai	1878	M.S Ratwara
Uttar Pradesh	9	Barabanki	129	Masauli	949	Lakshbar
Uttar Pradesh	9	Kanpur Nagar	157		0	Rasoolpur Umra
Uttar Pradesh	9	Pratapgarh	174		0	Chivila
Maharashtra	27	Bid	0	Beed	0	Nathapur
Karnataka	29	Tumakuru	548	Chiknayakanhalli	5905	Hoyasala Katti
Uttar Pradesh	9	Jaunpur	152		0	Belwan
Jharkhand	20	Deoghar	324	Sarath	3088	Upj M.S Kura
Andhra Pradesh	28	Prakasam	517	Kondapi	5476	Zphunethivanipalem
Telangana	36	Mahabubabad	688	Mahbubabad	5683	Zpps (B)Mahabubabad
Assam	18	Lakhimpur	295	Lakhimpur	2645	Lakhimpur
Tamil Nadu	33	Ariyalur	610	T. Palur	6260	Vikkoramanagalam
Chhattisgarh	22	Rajnandgaon	388		0	Bundeli
Maharashtra	27	Bid	0	Dharur	0	Arjandoha
Tamil Nadu	33	Dharmapuri	571	Pennagaram	6140	Pauparapatti
Karnataka	29	Bengaluru Rural	526	Devanahalli	5784	Vishwanathapura
Madhya Pradesh	23	Morena	417	Morena	3917	Hss Dathara (2302035606)
Uttar Pradesh	9	Agra	118	Jagner	871	Sigayach
Uttar Pradesh	9	Hardoi	150	Shahabad	1210	Uddharanpur

Table: Basic Details of Schools_Sample_Data (500 rows)

Search

Basic Details of Schools_S...

Affiliation Board for Higher S...

Affiliation Board for Seconda...

Avg Class Rooms Per School

Avg Enrollment Per School

Avg Students Per School

Avg Teachers Per School

Σ Class From (class_from)

Σ Class Rooms (class_rooms)

Σ Class To (class_to)

Cluster (cluster_name)

District (district_name)

Σ District Code (district_code)

Enrollment Time

Highest Enrollment

Σ Latitude (latitude)

Location Hierarchy

Location Type (location_type)

Σ Longitude (longitude)

Management (management)

Σ Non Primary Students (class_...

Σ Other Rooms (other_rooms)

Σ Pincode (pincode)

DATA MODELING

MODEL DESIGN

Purpose:

The data model was designed to support interactive analysis of school distribution, student enrollment, teacher availability, and infrastructure facilities across different states and districts. The model enables efficient reporting, filtering, and dashboard analysis.

MAIN TABLE

Basic Details of Schools Dataset

KEY FIELDS

School Name

State Name

District Name

Sub-District Name

Village Name

School Category

Management Type

Location Type

Total Students

Total Teachers

Class Rooms

Pre Primary Rooms

Other Rooms

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RELATIONSHIPS

Single Table Data Model

(No additional relationships were required because the dataset was maintained in a structured and consolidated format.)

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FILTER DIRECTION

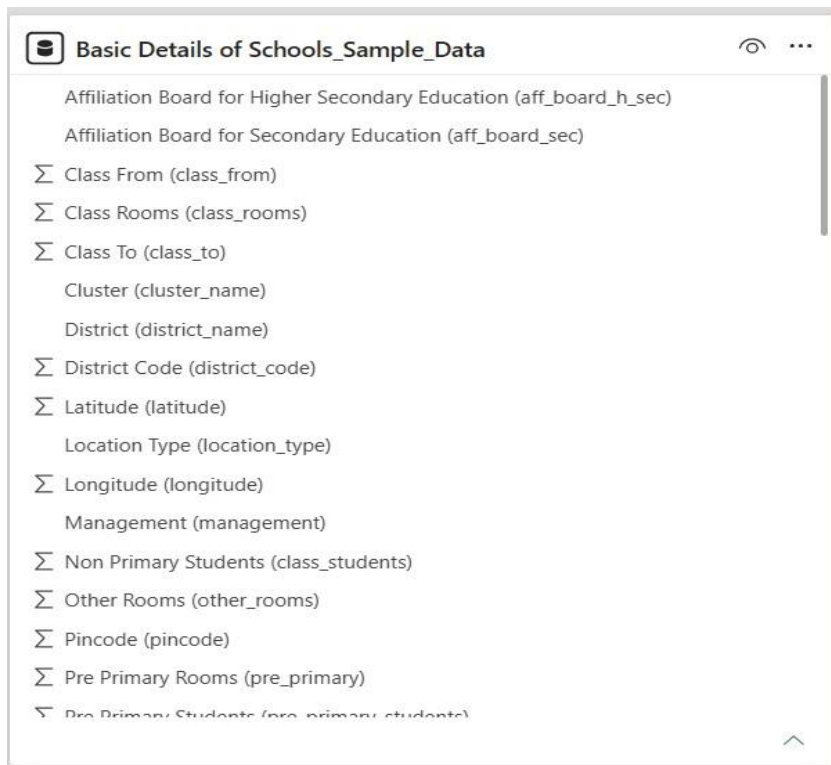
Single Direction

=====

MODEL BENEFITS

- ✓ Improved Report Performance
- ✓ Easy DAX Development
- ✓ Simplified Dashboard Design
- ✓ Efficient Data Analysis
- ✓ Interactive Filtering and Drill Analysis
- ✓ Better User Experience

SCREENSHOT



Single Table Data Model

DAX MEASURES USED

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TOTAL SCHOOLS

Total Schools =

DISTINCTCOUNT('Basic Details of Schools Sample_Data'[School Name
(school_name)])

Purpose:

Calculate the total number of schools.

=====

TOTAL STUDENTS

Total Students =

SUM('Basic Details of Schools_Sample_Data'[Total Students
(class_with_pre_primary_students)])

Purpose:

Calculate total student enrollment.

=====

TOTAL TEACHERS

Total Teachers =

SUM('Basic Details of Schools_Sample_Data'[Teachers (total_teachers)])

Purpose:

Calculate total teachers across all schools.

=====

TOTAL DISTRICTS

Total Districts =

DISTINCTCOUNT('Basic Details of Schools_Sample_Data'[District (district_name)])

Purpose:

Calculate the total number of districts.

=====

TOTAL STATES

Total States =

DISTINCTCOUNT('Basic Details of Schools_Sample_Data'[State (state_name)])

Purpose:

Calculate the total number of states.

=====

TOTAL CATEGORIES

Total Categories =

DISTINCTCOUNT('Basic Details of Schools_Sample_Data'[School Category (school_category)])

Purpose:

Calculate the total number of school categories.

=====

TOTAL MANAGEMENT TYPES

Total Management Types =

DISTINCTCOUNT('Basic Details of Schools_Sample_Data'[Management
(management)])

Purpose:

Calculate the total number of management types.

=====

TOTAL CLASS ROOMS

Total Class Rooms =

SUM('Basic Details of Schools_Sample_Data'[Class Rooms (class_rooms)])

Purpose:

Calculate total classrooms available.

=====

TOTAL PRE PRIMARY ROOMS

Total Pre Primary Rooms =

SUM('Basic Details of Schools_Sample_Data'[Pre Primary Rooms])

Purpose:

Calculate total pre-primary classrooms.

=====

TOTAL OTHER ROOMS

Total Other Rooms =

SUM('Basic Details of Schools_Sample_Data'[Other Rooms (other_rooms)])

Purpose:

Calculate total other rooms available

=====

AVERAGE STUDENTS PER SCHOOL

Avg Students Per School =

DIVIDE([Total Students],[Total Schools])

Purpose:

Calculate average students per school.

=====

AVERAGE TEACHERS PER SCHOOL

Avg Teachers Per School =

$\text{DIVIDE}([\text{Total Teachers}], [\text{Total Schools}])$

Purpose:

Calculate average teachers per school.

=====

AVERAGE ENROLLMENT PER SCHOOL

Avg Enrollment Per School =

$\text{DIVIDE}([\text{Total Enrollment}], [\text{Total Schools}])$

Purpose:

Calculate average enrollment per school.

=====

TEACHER STUDENT RATIO

Teacher Student Ratio =

$\text{DIVIDE}([\text{Total Students}], [\text{Total Teachers}])$

Purpose:

Calculate the average number of students per teacher.

VISUALIZATIONS USED

KPI CARDS

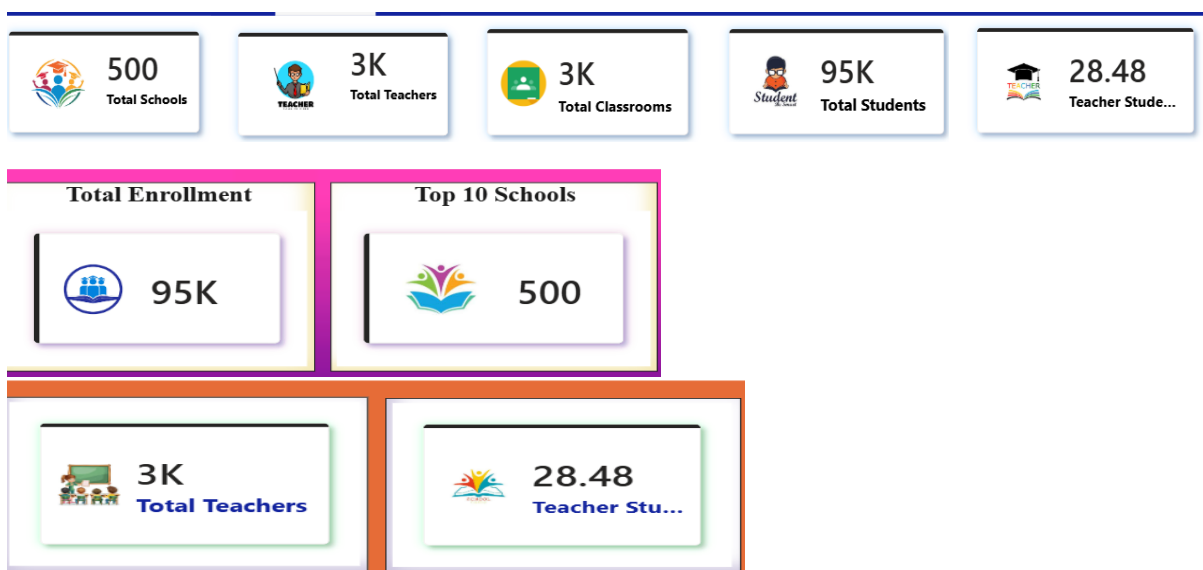
Purpose:

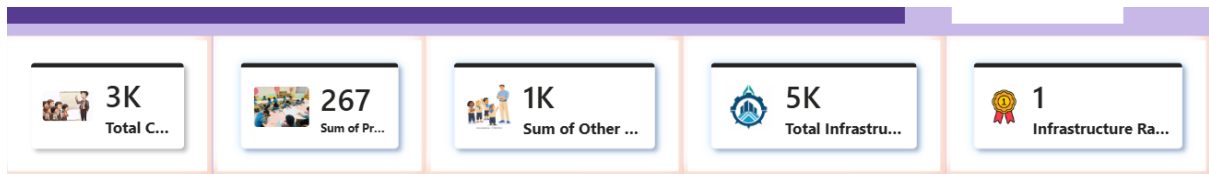
KPI Cards are used to display key educational metrics and provide a quick overview of school, enrollment, teacher, and infrastructure statistics.

KPIs Included:

- Total Schools
- Total Students
- Total Teachers
- Total Districts
- Total States
- Total Categories
- Total Management Types
- Total Class Rooms
- Total Other Rooms
- Teacher Student Ratio
- Highest Enrollment

SCREENSHOT





CLUSTERED BAR CHART

Purpose:

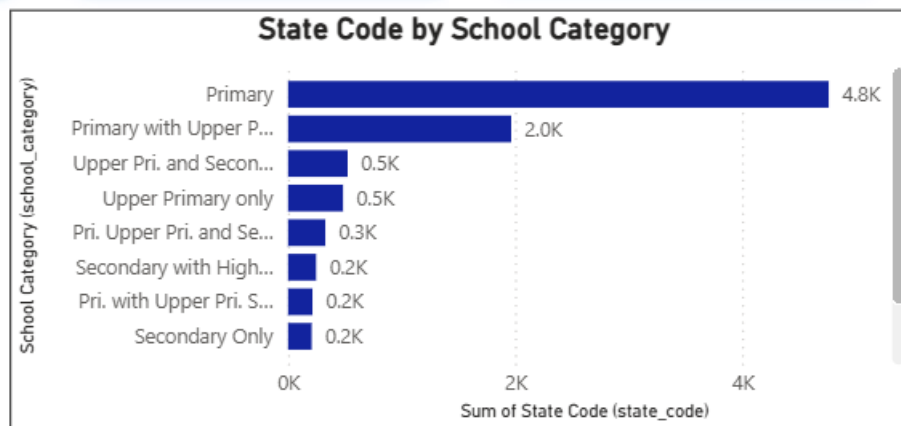
Clustered Bar Charts are used to compare educational metrics across states, districts, and school categories.

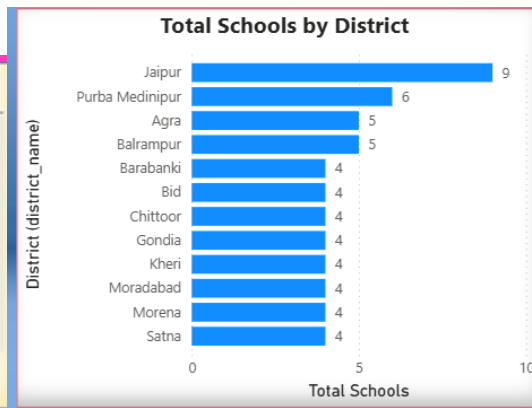
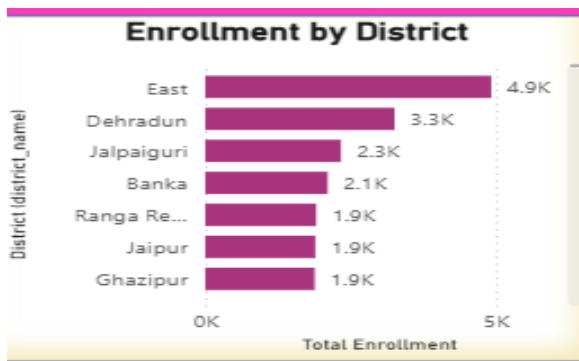
The charts were used to analyze:

- State code by school category
- Enrollment by districts
- Total schools by districts

These visualizations help identify top-performing and low-performing regions.

SCREENSHOT





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CLUSTERED COLUMN CHART

Purpose:

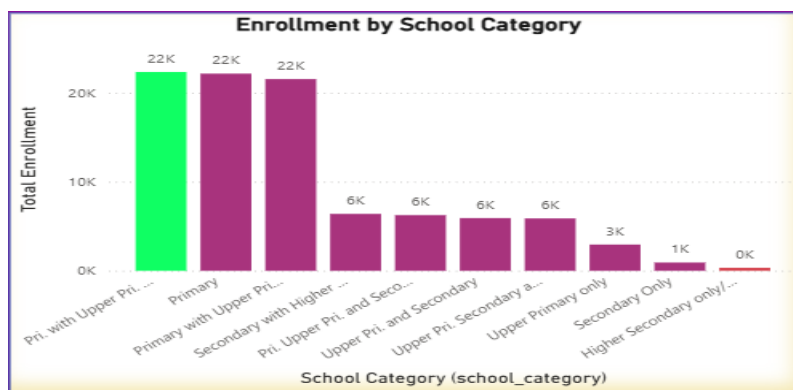
Clustered Column Charts are used to compare distributions and rankings across categories.

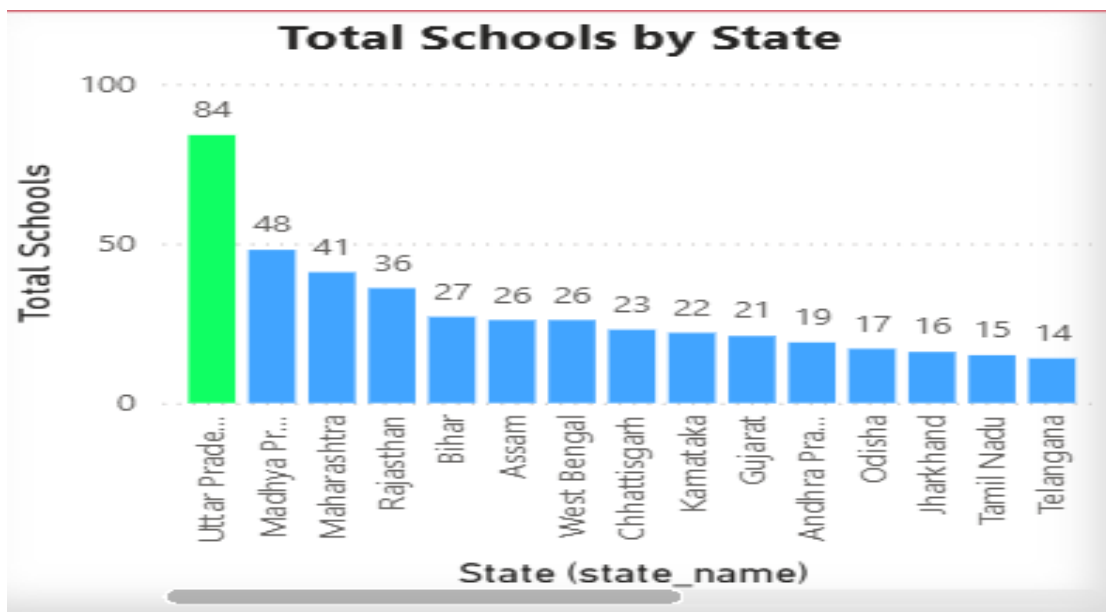
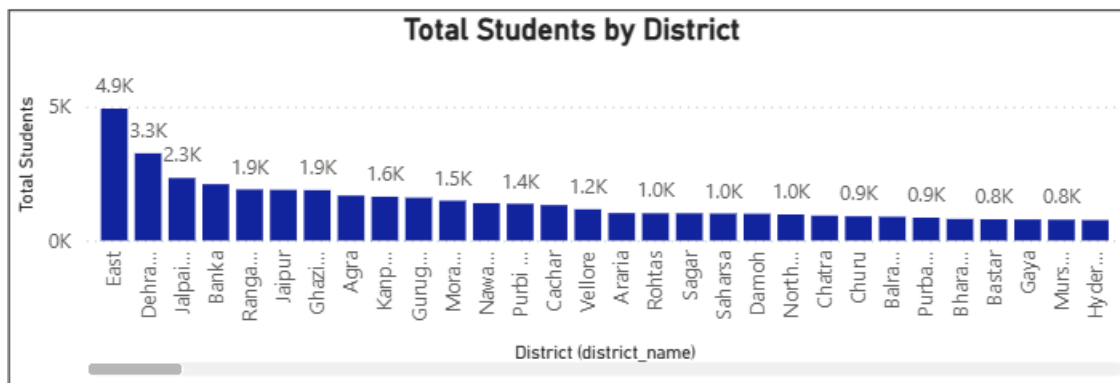
The charts were used to analyze:

- Enrollment by School Category
- Students by districts and states

These visualizations help compare educational performance across categories.

SCREENSHOT





DONUT CHART & PIE CHART

Purpose:

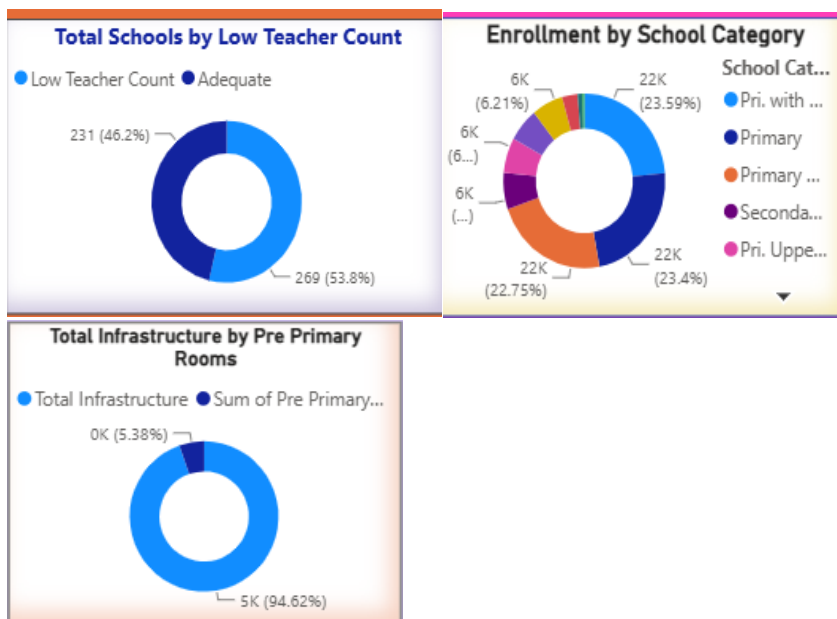
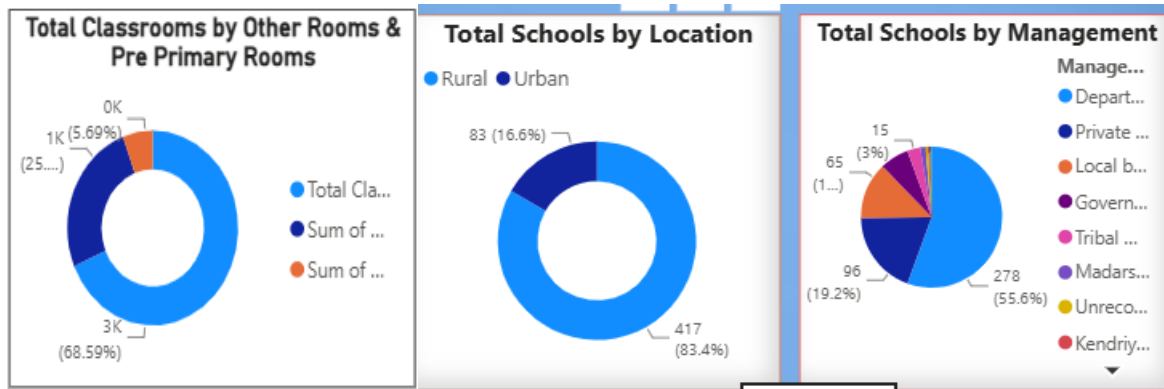
Charts are used to show percentage distribution and category-wise contribution.

The charts were used to analyze:

- Schools by Location
- Schools by Management Type
- Schools by low teacher count
- Enrollment by school category
- Infrastructure by Pre Primary rooms

These visuals help identify dominant categories and management types.

SCREENSHOT



TREEMAP

Purpose:

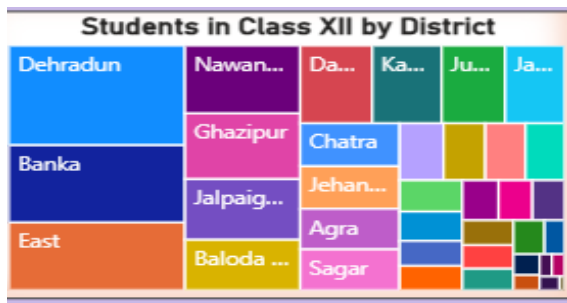
Treemaps are used to visualize hierarchical data and compare classroom availability across districts.

The charts were used to analyze:

- Students in Class XII by District

These visualizations help identify districts with higher and lower infrastructure availability.

SCREENSHOT



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MAP VISUALIZATION

Purpose:

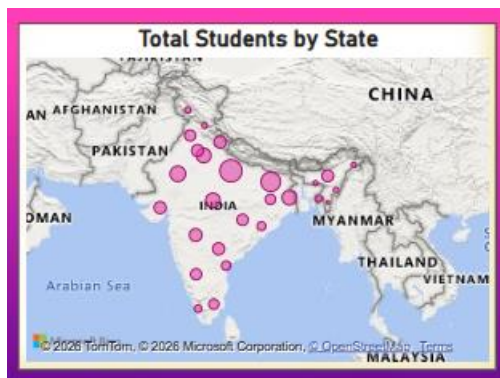
Map Visualizations are used to display geographical distribution of schools.

The charts were used to analyze:

- Total Students by State

These visuals provide location-based insights and support geographical analysis.

SCREENSHOT



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LINE CHART

Purpose:

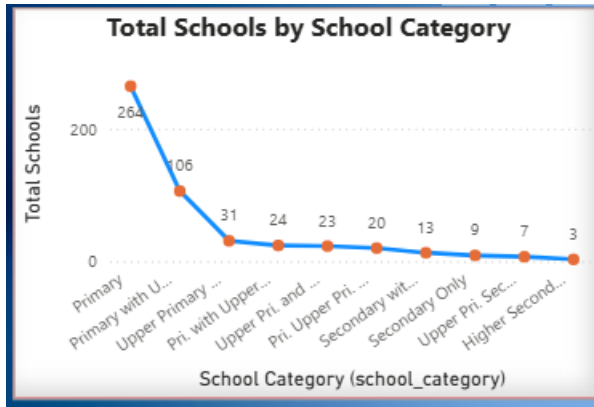
Used to analyze the distribution of schools across different school categories.

The charts were used to analyze:

- Total schools by school category

These visuals help identify which school categories have the highest and lowest number of schools, supporting educational planning and resource allocation.

SCREENSHOT



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TABLE / MATRIX

Purpose:

Used to display detailed school level information such as village wise student and teacher counts for analysis and decision making.

The visual was used to analyze:

- Total Students by village
- Total Teachers by village

These visuals provide detailed records and support decision-making.

SCREENSHOT

Village (village_name)	Total Students	Total Teachers
Zari Bzr.	47	2
Yadadore	179	6
Ward-5	235	10
Ward-45	197	7
Ward-4	35	4
Ward-28	109	6
Ward-21	518	19
Ward No-2	66	2
Ward No.56	48	3
Ward No.-3	537	15
Ward No.-23	110	1
Ward No.-16	45	1
Ward No. 25	100	10
Total	94680	3324

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SLICERS

Purpose:

Slicers provide interactive filtering and dynamic analysis across all dashboards.

Slicers Implemented:**State Slicer**

- Enables state-wise analysis.

District Slicer

- Enables district-wise analysis.

School Category Slicer

- Filters schools by category.

Management Type Slicer

- Enables management-wise analysis.

All slicers are synchronized across dashboards using Sync Slicers.

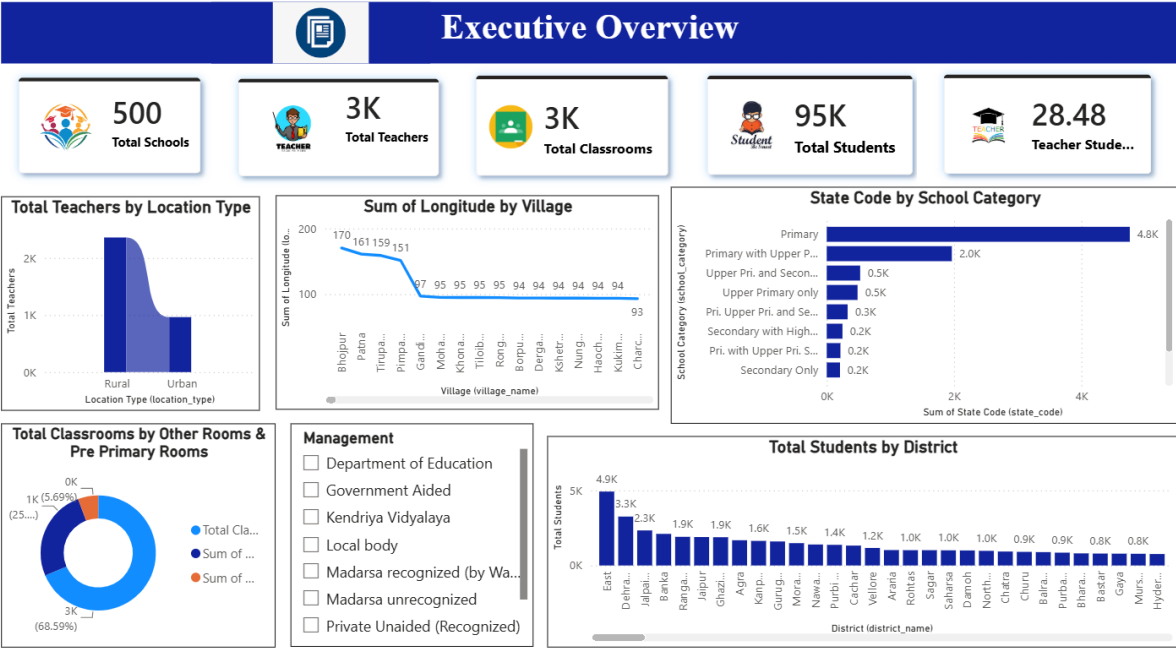
SCREENSHOT

State	District	School Category	Management
All	All	All	All

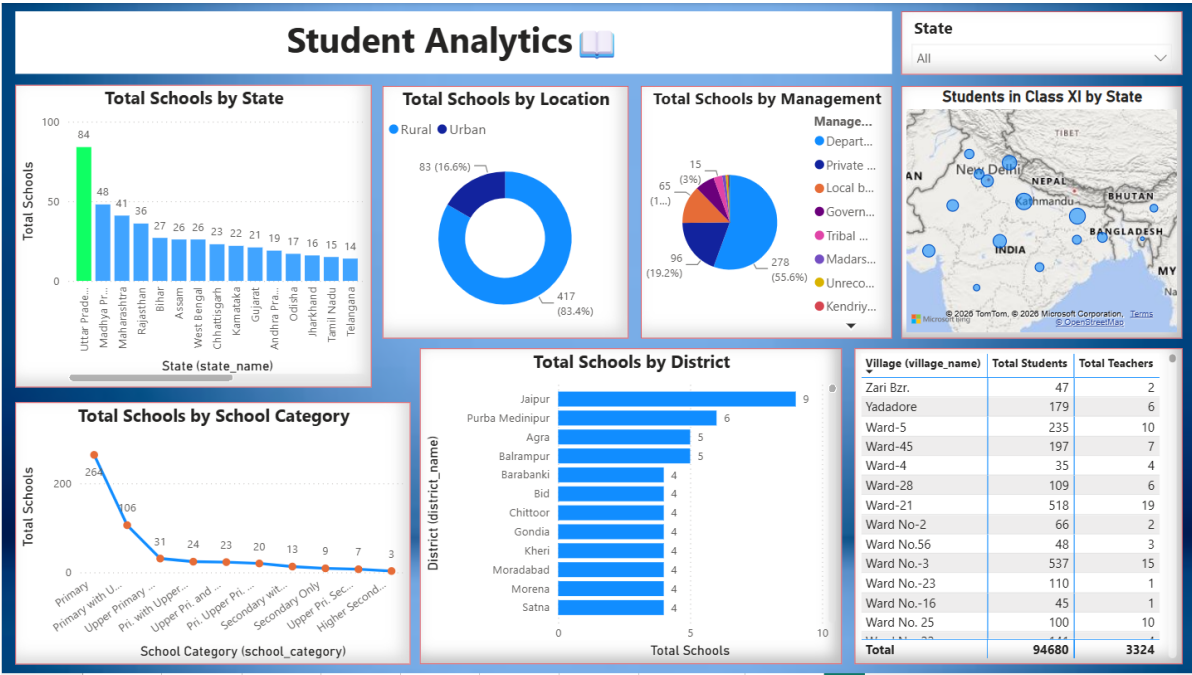
DASHBOARD SCREENSHOTS

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EXECUTIVE OVERVIEW



SCHOOL ANALYTICS



Explanation

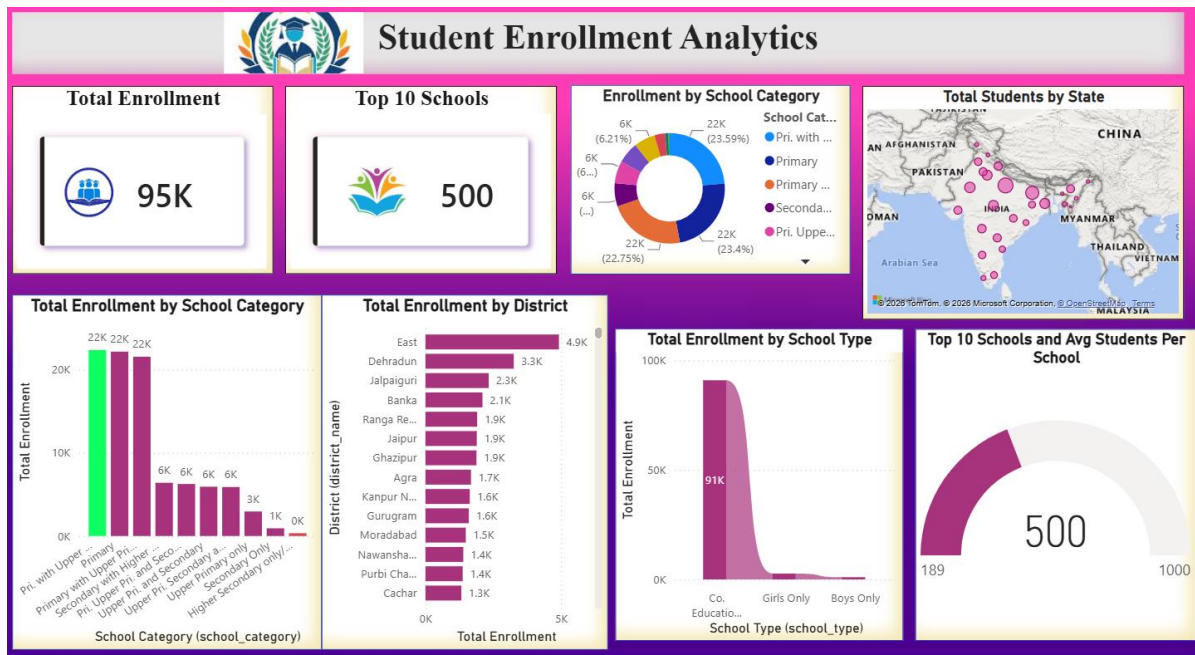
The School Analytics Dashboard focuses on analyzing the distribution of schools across states, districts, school categories, management types, and location types. It helps identify the concentration of schools and provides insights into educational infrastructure distribution across regions.

Key Insights:

- Uttar Pradesh has the highest number of schools in the dataset.
- Primary schools account for the largest share of schools.
- Government-managed schools form the majority of institutions.
- Rural schools outnumber urban schools.
- School distribution varies considerably across districts.

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ENROLLMENT ANALYTICS



Explanation

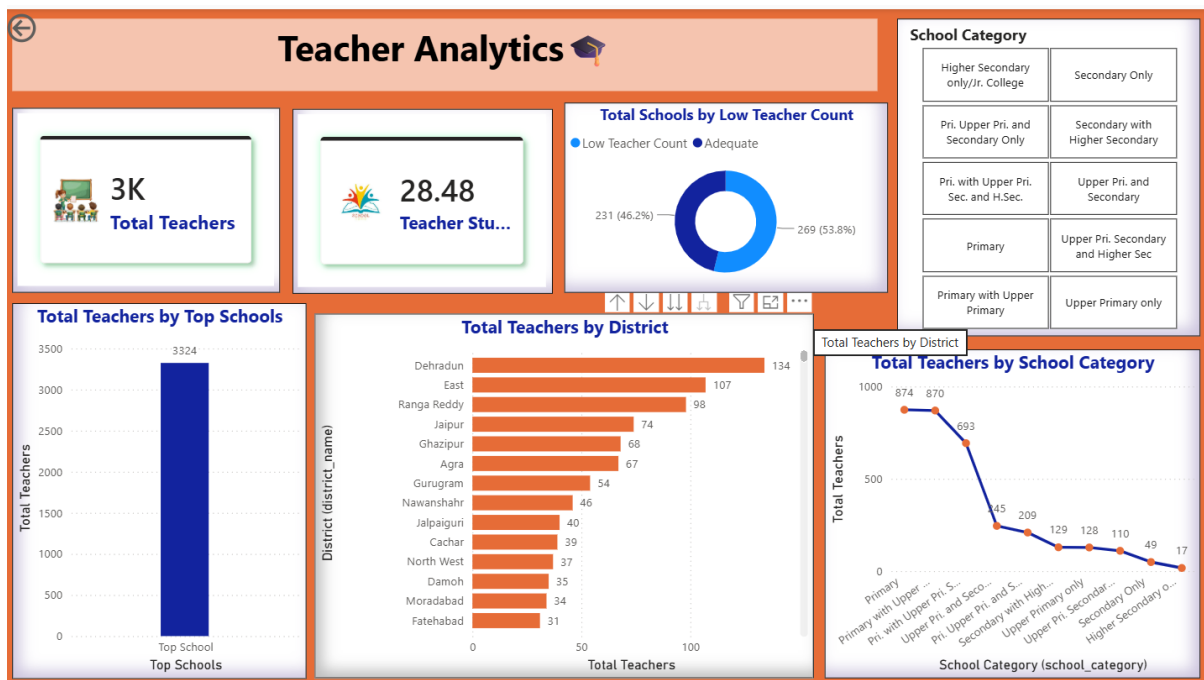
The Enrollment Analytics Dashboard focuses on student enrollment patterns across states, districts, school categories, and school types. It helps identify regions with high and low enrollment and supports educational resource planning.

Key Insights:

- Total Student Enrollment: 95K
- Top 10 Schools: 500
- Enrollment differs significantly across districts.
- Certain school categories contribute a larger share of total enrollment.
- State-wise analysis highlights enrollment concentration areas.
- Interactive filters allow detailed enrollment comparisons.

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TEACHER ANALYTICS



Explanation

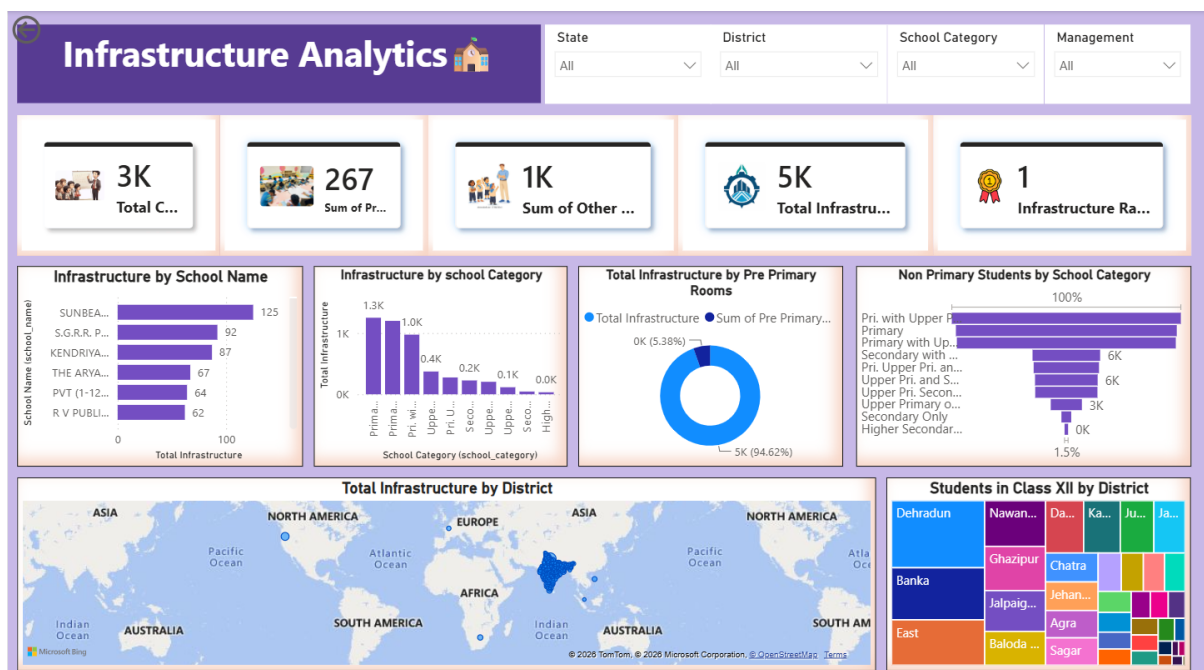
The Teacher Analytics Dashboard evaluates teacher availability and allocation across schools, districts, and states. It provides insights into teacher distribution and identifies schools requiring additional teacher support.

Key Insights:

- Total Teachers: 3K
- Teacher Student Ratio: 28.48
- Teacher distribution varies across districts.
- Several schools have low teacher counts and may require additional allocation.
- Conditional formatting highlights schools requiring attention.
- Interactive filtering supports detailed teacher analysis.

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INFRASTRUCTURE ANALYTICS



Explanation

The Infrastructure Analytics Dashboard focuses on classroom availability and infrastructure facilities across schools. It helps identify infrastructure gaps and supports planning for educational facility improvements.

Key Insights:

- Total Class Rooms: 3K
- Total Pre Primary Rooms: 267
- Total Other Rooms: 1K
- Total Infrastructure : 5K
- Infrastructure Rank : 1
- A few schools have significantly higher infrastructure availability than others,
- Primary and Upper Primary schools account for the highest infrastructure.
- Infrastructure distribution is uneven across districts.
- Interactive filtering enables detailed infrastructure assessment.

KEY INSIGHTS

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INSIGHT-1

The dataset contains approximately 500 schools, 95K students, and 3K teachers, providing a comprehensive view of educational infrastructure and enrollment patterns.

INSIGHT-2

Primary schools constitute the largest share of schools, indicating that primary education forms the foundation of the educational system.

INSIGHT-3

Rural schools significantly outnumber urban schools, highlighting the importance of rural educational development and resource allocation.

INSIGHT-4

Student enrollment is unevenly distributed across districts, with certain districts accounting for a larger proportion of enrolled students.

INSIGHT-5

The average Teacher-Student Ratio is approximately 28.48, helping assess teacher workload and educational quality across schools.

INSIGHT-6

Several schools have low teacher counts and require additional teacher allocation to maintain balanced educational delivery and improve learning outcomes.

INSIGHT-7

Classroom availability varies considerably across states and districts, indicating infrastructure disparities that may affect teaching and learning environments. .

INSIGHT-8

Government-managed schools represent the majority of institutions, demonstrating their significant role in delivering education across regions.

INSIGHT-9

Infrastructure analysis identifies schools with poor classroom availability, helping administrators prioritize investments and facility improvements.

INSIGHT-10

Interactive analysis through slicers, drill-through, tooltips, and dynamic dashboards enables efficient exploration of educational trends and supports data-driven decisionmaking.

BUSINESS RECOMMENDATIONS

RECOMMENDATION-1

Educational authorities should prioritize districts with low school availability to ensure equal access to education across all regions.

RECOMMENDATION-2

Schools with high student enrollment and low teacher availability should receive additional teacher allocations to improve the Teacher-Student Ratio.

RECOMMENDATION-3

Government agencies should focus on improving infrastructure in schools with limited classroom facilities and poor infrastructure conditions.

RECOMMENDATION-4

Rural schools should receive targeted investments to strengthen educational facilities and ensure equitable learning opportunities.

RECOMMENDATION-5

Regular monitoring of enrollment trends, teacher distribution, and infrastructure availability should be conducted to support effective educational planning.

FEATURES IMPLEMENTED

- ✓ KPI Cards
- ✓ Drill Down
- ✓ Drill Through
- ✓ Custom Tooltips
- ✓ Dynamic Titles
- ✓ Slicers
- ✓ Sync Slicers
- ✓ Conditional Formatting
- ✓ Navigation Buttons

- ✓ Interactive Maps
- ✓ Treemap Visualization
- ✓ Cross Filtering
- ✓ Custom Theme

CHALLENGES FACED

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Data Cleaning Challenges
Handling Missing Values
DAX Measure Development
Dynamic Title Implementation
Drill Through Configuration
Tooltip Design
Dashboard Layout Design
Visual Selection and Optimization
Performance Tuning

LEARNINGS

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Power Query
Data Cleaning
Data Modeling
DAX Development
Dashboard Design
Interactive Reporting
Storytelling with Data
Business Analysis

PROJECT STORYTELLING

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BUSINESS SCENARIO

Educational departments and policymakers manage large volumes of school-related data including school distribution, student enrollment, teacher availability, and infrastructure facilities. However, extracting meaningful insights from raw data is challenging.

This project transforms educational data into interactive dashboards that support monitoring, analysis, and decision-making for educational planning and resource allocation.

BUSINESS PROBLEM

- Which states and districts have the highest and lowest number of schools?
- How is student enrollment distributed across schools and districts?
- Are teachers adequately distributed across schools?
- Which schools require additional teacher allocation?
- How is infrastructure distributed across different regions?
- Which schools require infrastructure improvements?
- How can educational resources be allocated more effectively?

ANALYSIS APPROACH

- Analyzed school distribution across states and districts.
- Evaluated student enrollment patterns.
- Assessed teacher availability and Teacher-Student Ratios.
- Analyzed classroom and infrastructure availability.
- Compared school categories and management types.
- Identified schools requiring additional support.
- Implemented Drill Down, Drill Through, Tooltips, Dynamic Titles, and Conditional Formatting.
- Developed interactive dashboards for educational decision-making.

KEY FINDINGS

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The dataset contains approximately 500 schools, 95K students, and 3K teachers.

- Primary schools constitute the largest share of educational institutions.
- Rural schools outnumber urban schools.
- Student enrollment is concentrated in a limited number of districts.
- Teacher availability varies significantly across districts.
- Several schools require additional teacher allocation.
- Infrastructure availability differs across states and districts.
- Schools with limited classroom facilities require infrastructure improvements.
- Government-managed schools represent the majority of institutions.
- Interactive dashboard analysis improves visibility into educational performance and resource allocation.

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BUSINESS IMPACT

- =====
- Supports data-driven educational planning.
 - Improves monitoring of student enrollment trends.
 - Helps identify teacher shortages.
 - Assists in infrastructure planning and development.
 - Enables efficient allocation of educational resources.
 - Improves decision-making for administrators and policymakers.
 - Provides a centralized educational analytics platform.

PROJECT DEPLOYMENT

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DEPLOYMENT OBJECTIVE

Deploy the Power BI solution to provide stakeholders with access to interactive educational dashboards and real-time insights.

DEPLOYMENT PLATFORM

Power BI Service

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DEPLOYMENT PROCESS

STEP-1

Validate Data Model and DAX Measures in Power BI Desktop.

STEP-2

Optimize Dashboard Layout and Performance.

STEP-3

Publish PBIX File to Power BI Service.

STEP-4

Configure Dataset Refresh Settings.

STEP-5

Create Dashboards from Published Reports.

STEP-6

Configure User Access and Security Settings.

STEP-7

Share Dashboards with Stakeholders.

BUSINESS BENEFITS

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- Centralized Educational Analytics Platform
- Interactive Dashboard Reporting
- Improved Resource Allocation
- Better Infrastructure Monitoring
- Enhanced Enrollment Analysis
- Improved Teacher Allocation Planning
- Faster Data-Driven Decisions

CONCLUSION

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The School Infrastructure & Enrollment Analytics Dashboard successfully transformed educational data into interactive and insightful visualizations. The project analyzed school distribution, student enrollment, teacher availability, and infrastructure facilities across multiple regions.

Through the use of Power BI features such as KPI Cards, Drill Down, Drill Through, Tooltips, Dynamic Titles, Conditional Formatting, and Interactive Dashboards, the project provides valuable insights for administrators, policymakers, and educational institutions.

The solution supports informed decision-making, efficient resource allocation, and improved educational planning through comprehensive data analysis and reporting.